



Cerberus WAF: Advanced Web Application Firewall for Modern Cybersecurity Challenges.

Supervised by:

Dr. Ali Elsherbeni Takieldeem

Team Members:

Name : Esraa salama Mohammed	ID : 4221177
Name : Fathy Abdelhamed Fathy	ID : 4221246
Name : Malak Ali Eldawansy	ID : 4221187
Name : Menna Essam Abdelhamid	ID : 4221272
Name : Mariam Mahmoud Ayoub	ID : 4221294
Name : Mohamed Hisham Salah El-din	ID : 4221340
Name : Naniss ElSayed Yahia	ID : 4221415

CERERUSWAF

WEB APPLICATION FIREWALL



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Chapter 1

Introduction

In the digital era, the internet has become the backbone of commercial activity, governmental operations, and personal communication. Web applications process billions of transactions daily, handling sensitive financial records, personal identification data, healthcare information, and confidential business communications. This explosion of web-based activity has made these applications prime targets for malicious actors seeking unauthorized access, data exfiltration, service disruption, and financial gain.

The scale of this exposure is difficult to overstate. Industry breach reports consistently show that web applications and APIs represent the single largest attack surface for organizations of every size, and the financial consequences of a successful compromise extend far beyond the immediate incident response cost. Regulatory fines under frameworks such as GDPR and PCI DSS, loss of customer trust, reputational damage, and the operational disruption of remediation efforts all compound the direct cost of a breach. For organizations operating in regulated industries such as finance, healthcare, and government services, a single successful attack can trigger cascading compliance failures with consequences that persist for years.

Compounding this problem is the shift toward API-first architectures and microservices, which dramatically expand the number of distinct endpoints that must be secured. Where a traditional monolithic web application might expose a handful of entry points, a modern microservices-based system can expose hundreds or thousands of API endpoints, each representing a potential point of compromise. This proliferation of attack surface has outpaced the ability of many security teams to manually review and harden each endpoint, making automated, intelligent protection mechanisms not merely beneficial but essential.

The threat landscape facing modern web applications is both vast and constantly evolving. Attackers employ increasingly sophisticated techniques to exploit vulnerabilities, ranging from classic injection attacks and cross-site scripting to advanced zero-day exploits and API-specific threats. Traditional security measures such as network firewalls and intrusion detection systems operate at lower network layers and are fundamentally ill-equipped to understand or inspect application-layer traffic at the semantic level required to detect modern attacks.

This layer mismatch is not a minor technical detail — it represents a fundamental architectural gap in the security posture of most organizations. A network firewall configured to allow traffic on port 443 cannot distinguish between a legitimate HTTPS request to a login endpoint and a

carefully crafted SQL injection payload directed at the same endpoint, because both arrive as encrypted TCP packets destined for the same port. Only by terminating the TLS connection and inspecting the decrypted HTTP request content — including the URL path, query string, headers, cookies, and body — can a security control make an informed decision about whether the request is legitimate. This is precisely the capability that a Web Application Firewall provides, and its absence in conventional network security infrastructure is the primary motivation for WAF adoption.

This reality has made Web Application Firewalls (WAFs) an essential component of any comprehensive cybersecurity strategy. A WAF sits as a reverse proxy between clients and web application servers, inspecting, analyzing, and filtering HTTP/HTTPS traffic to identify and block malicious requests before they reach vulnerable application code. However, the majority of commercial WAF solutions are prohibitively expensive for small and medium-sized organizations, while free alternatives often suffer from high false positive rates, limited detection capabilities, or complex configuration requirements.

This creates a troubling bifurcation in the security landscape: large enterprises with substantial security budgets can afford best-in-class commercial WAF platforms with dedicated support teams, continuous rule updates, and managed threat intelligence feeds, while small and medium-sized businesses — which collectively represent the vast majority of internet-facing web applications — are left to choose between expensive solutions they cannot afford, free solutions that require expertise they do not have, or no WAF protection at all. Given that attackers frequently target smaller organizations precisely because they are perceived as softer targets with weaker defenses, this gap represents a systemic vulnerability in the broader internet ecosystem, not merely an inconvenience for individual organizations.

Cerberus WAF addresses this challenge by providing an industrial-grade, freely available web application firewall and API security gateway that leverages cutting-edge artificial intelligence, machine learning, and semantic analysis technologies. Named after the mythological three-headed guardian of the underworld, Cerberus WAF embodies its namesake's vigilant, multi-layered protection — defending web infrastructure at the traffic layer, the system layer, and the application runtime layer simultaneously.

1.1 Overview

Cerberus WAF is an open-source, high-performance Web Application Firewall and API Security Gateway that operates as a cloud-based reverse proxy. It is designed to protect web applications from the full spectrum of OWASP Top 10 threats and beyond, combining multiple complementary detection and prevention technologies into a unified, easy-to-deploy solution.

At its architectural core, Cerberus WAF is built upon the battle-tested Nginx web server enhanced with LuaJIT scripting capabilities, providing the dual benefits of high throughput processing and flexible, programmable security logic. This architecture enables Cerberus WAF to handle enterprise-level traffic volumes while maintaining sub-millisecond latency overhead per request.

The choice of Nginx as the foundational HTTP processing engine was deliberate and informed by its proven track record in some of the highest-traffic environments on the internet. Nginx's event-driven, non-blocking architecture allows a single worker process to handle thousands of concurrent connections using a small, fixed memory footprint, in contrast to thread-per-connection models that scale poorly under high concurrency. By embedding the security logic directly into Nginx's request processing lifecycle via LuaJIT — rather than implementing it as a separate proxy or sidecar service — Cerberus WAF avoids the additional network hops and serialization overhead that would otherwise be incurred, which is a key factor in its ability to maintain low latency even when performing computationally intensive operations such as semantic parsing and machine learning inference on every request.

The system implements a three-layer defense model that distinguishes it from conventional WAF solutions. At the traffic layer, all incoming HTTP and HTTPS requests pass through a comprehensive inspection pipeline consisting of semantic detection engines, machine learning classifiers, rule-based filters, and access control mechanisms. At the system layer, a Host Intrusion Prevention System (HIPS) monitors and controls kernel-level activities such as process creation, file system operations, network communications, and privilege escalation attempts. At

the runtime layer, a Runtime Application Self-Protection (RASP) module is injected directly into application execution environments including Java Virtual Machines (JVM) and PHP Zend engines, providing real-time, context-aware threat interception.

The management interface provides administrators with a clean, intuitive web-based dashboard for configuring sites, managing SSL certificates, reviewing attack logs, adjusting rule sets, and monitoring system performance — all without requiring deep command-line expertise.

This design philosophy reflects a broader principle that underlies the entire Cerberus WAF project: security tooling should not require its operators to be security experts in order to derive meaningful protective value. While advanced users retain full access to the underlying Lua scripting environment and configuration files for fine-grained customization, the default experience is designed around guided workflows, sensible defaults, and clear visual feedback that allow a system administrator with general IT skills — but no specialized security training — to deploy and operate an effective WAF. This accessibility is itself a security contribution, because a powerful tool that is too complex to configure correctly often provides less real-world protection than a simpler tool that is configured correctly from the outset.

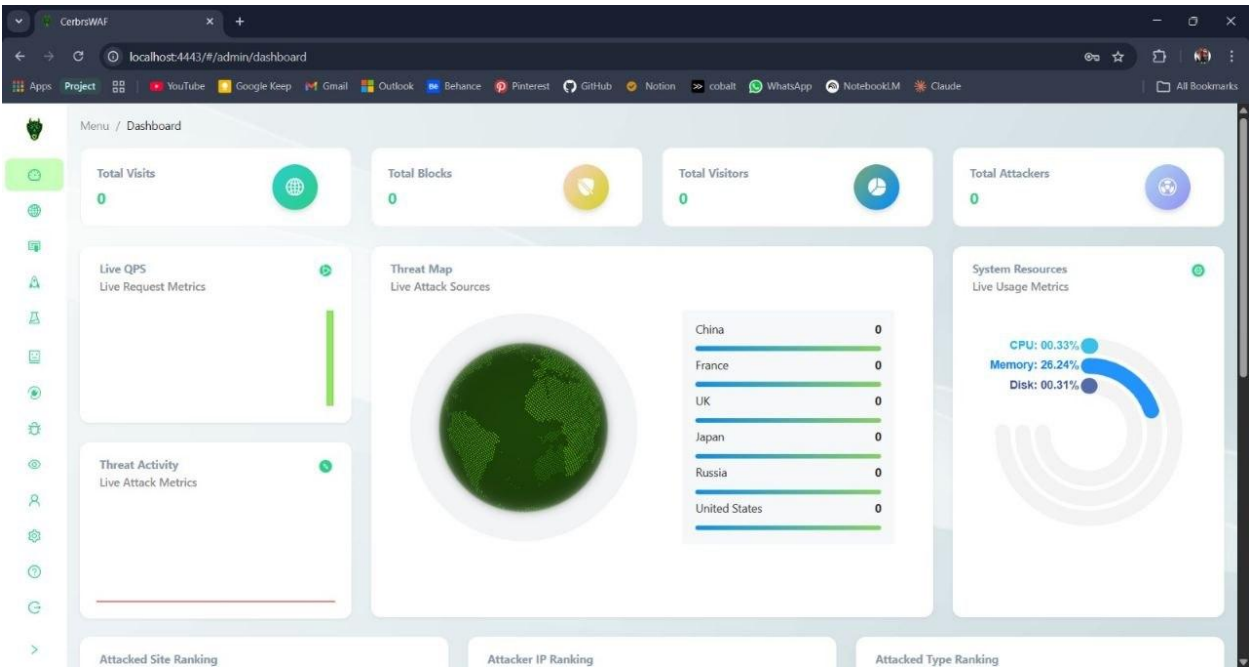


Figure 1.1: Main Cerberus WAF Management Dashboard

1.2 Goals

The primary goals of the Cerberus WAF project are defined across four strategic dimensions: security effectiveness, operational performance, ease of use, and extensibility.

From a security perspective, Cerberus WAF aims to achieve industry-leading detection rates against both known and unknown (zero-day) attacks. The system targets a false positive rate below 0.1% to ensure legitimate traffic is not disrupted, a detection accuracy above 99% against common web attack vectors, and zero-day threat mitigation through behavioral and machine learning approaches that do not rely solely on signature databases.

Operationally, the system is designed to introduce minimal latency overhead to protected applications, operate reliably at high traffic volumes without becoming a bottleneck, and integrate seamlessly with existing infrastructure through standard reverse proxy deployment patterns.

In terms of usability, Cerberus WAF is designed to be deployable within minutes using a single installation command, manageable through an intuitive web dashboard without requiring security expertise, and self-maintaining with automated SSL certificate renewal and background rule updates.

For extensibility, the system provides a powerful Lua-based scripting API allowing advanced administrators to write custom detection and response logic, a plugin architecture enabling third-party feature extensions, and a rules API supporting dynamic rule management without system restarts.

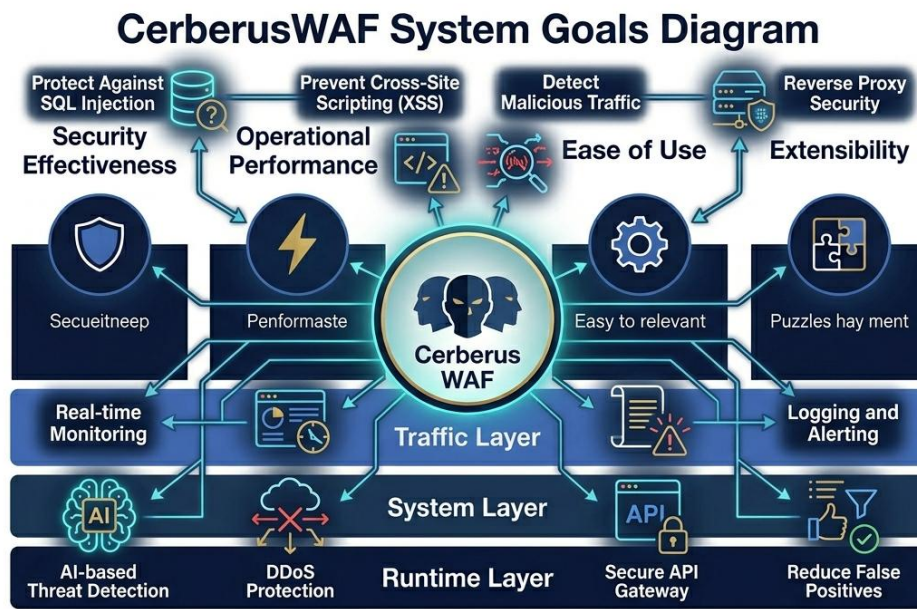


Figure 1.2: Cerberus WAF Strategic Goals and Architecture Overview

These four dimensions — security effectiveness, operational performance, ease of use, and extensibility — were not selected arbitrarily. They correspond directly to the four most common reasons that WAF deployments fail to deliver their intended protective value in practice. A WAF that detects attacks effectively but introduces unacceptable latency will be disabled by operations teams under load. A WAF that performs well but generates excessive false positives will be configured into a permissive, ineffective mode to stop legitimate traffic from being blocked. A WAF that is effective and performant but too complex to configure correctly will be deployed with default settings that fail to address the organization’s specific risk profile. And a WAF that excels in the first three dimensions but cannot be extended or customized will eventually become a constraint on the organization as its needs evolve. By explicitly designing for all four dimensions simultaneously, Cerberus WAF aims to avoid each of these common failure modes.

It is worth noting that these goals are validated through the empirical testing presented in Chapter 4, which demonstrates that the project’s targets for detection accuracy, false positive rate, and performance overhead were not merely aspirational but were achieved and measured against a substantial test corpus of over 33,000 samples. This evidence-based approach to goal-setting and validation distinguishes the project from purely theoretical proposals and grounds its claims in reproducible, quantitative results.

1.3 Problem Statement

The protection of web applications and APIs against malicious traffic represents one of the most pressing and complex challenges in contemporary information security. Despite widespread awareness of web-based threats, organizations of all sizes continue to suffer devastating breaches, data theft, service outages, and financial losses attributable to vulnerabilities in their web-facing infrastructure.

Several interconnected problems define the current state of web application security:

The first is the inadequacy of traditional security measures. Conventional network firewalls and intrusion detection systems operate at the network and transport layers (OSI Layers 3 and 4) and lack the capability to interpret or inspect application-layer (Layer 7) HTTP traffic. They cannot distinguish between a legitimate database query submitted by an authorized user and a SQL injection attack embedded within a similar-looking HTTP request. This fundamental limitation renders them insufficient for protecting web applications.

To illustrate this limitation concretely, consider a typical e-commerce checkout endpoint that accepts a product identifier as a URL parameter. A legitimate request might include a parameter such as `product_id=4521`, while an attacker attempting a SQL injection might submit `product_id=4521 OR 1=1--`. From the perspective of a network firewall operating at Layers 3 and 4, both requests are simply TCP packets carrying an HTTP GET request to the same destination IP and port, with the same protocol headers — there is nothing at the network level that distinguishes the malicious request from the benign one. Only application-layer inspection, capable of parsing the HTTP request and evaluating the semantic content of its parameters, can identify the second request as an injection attempt.

The second problem is the high cost and complexity of effective WAF solutions. Commercial WAF products from leading vendors are expensive, typically requiring licensing fees in the range of tens of thousands of dollars annually, placing them beyond reach for small and medium enterprises. Open-source alternatives such as ModSecurity, while capable, require significant expertise to configure correctly and suffer from high false positive rates when deployed with permissive rule sets.

The expertise barrier associated with tools such as ModSecurity is substantial in practice. Configuring the OWASP Core Rule Set to an appropriate sensitivity level for a specific

application typically requires an iterative tuning process during which administrators must review logs, identify rules that are generating false positives for legitimate application behavior, and create application-specific exceptions — a process that can take weeks of dedicated effort and ongoing maintenance as the application evolves. Many organizations that deploy ModSecurity either never complete this tuning process, leaving the WAF in a permissive 'detection-only' mode that provides no actual blocking protection, or they disable problematic rule categories entirely, significantly narrowing the effective protection coverage.

The third challenge is the rapidly evolving nature of web attacks. The attack surface of modern web applications is enormous and constantly changing. Novel attack techniques, previously unknown vulnerabilities (zero-days), and polymorphic payloads specifically crafted to evade signature-based detection systems render purely rule-based WAFs increasingly inadequate. Security teams face the impossible task of manually updating rule sets fast enough to keep pace with emerging threats.

The scale of this challenge is illustrated by the volume of new vulnerabilities disclosed each year. The National Vulnerability Database routinely records tens of thousands of new CVE entries annually, and a meaningful subset of these are remotely exploitable web application vulnerabilities. Even an extremely well-resourced security team cannot realistically review every disclosure, assess its relevance to their specific application stack, develop a corresponding WAF rule, test that rule for false positives, and deploy it — all before attackers begin scanning the internet for vulnerable, unpatched systems, which research consistently shows often begins within hours of public disclosure.

The fourth problem is the gap between traffic-layer and application-layer defense. Even a perfectly configured WAF can be bypassed through application-level techniques. Vulnerabilities in the application runtime environment, malicious file writes to the server filesystem, unauthorized process execution, and privilege escalation attacks can all occur in ways that bypass traffic-layer inspection entirely, leaving organizations exposed despite having a WAF in place.

Cerberus WAF addresses all four of these problems through its multi-layered architecture, AI-enhanced detection engines, zero-day behavioral analysis, free and open licensing, and integrated HIPS and RASP capabilities.

Each of these four problems maps to a specific architectural element of the proposed solution, which is described in full detail in Chapter 3. The inadequacy of traditional network-layer security is addressed by Cerberus WAF's deep application-layer inspection pipeline. The cost and complexity barrier is addressed by the project's free, open-source licensing model and one-command deployment process. The rapidly evolving threat landscape is addressed by the machine learning-based anomaly detection subsystem, which does not depend on manually maintained signatures. And the gap between traffic-layer and application-layer defense is addressed by the integration of HIPS and RASP, which extend protection beyond the network perimeter into the operating system and application runtime itself.

1.4 Contribution

This project makes several significant contributions to the field of web application security:

Technical Innovation: Cerberus WAF represents a first-of-its-kind integration of traffic-layer semantic analysis, system-layer host intrusion prevention, and runtime application self-protection within a single, cohesive, freely available platform. This three-layer defense model provides comprehensive protection that no existing free solution matches.

The significance of this integration becomes clear when considering how attacks unfold in practice. A sophisticated attack rarely consists of a single malicious request that a traffic-layer WAF either blocks or fails to block. Instead, real-world compromises often involve a sequence of steps: initial reconnaissance to identify a vulnerability, an exploitation attempt that may partially succeed despite traffic-layer defenses, an attempt to establish persistence or execute commands on the compromised system, and finally an attempt to exfiltrate data or pivot to other systems. A traffic-layer WAF alone can only intervene at the first stage. By providing system-layer monitoring through HIPS and runtime-layer monitoring through RASP, Cerberus WAF retains the ability to detect and interrupt an attack even if the initial traffic-layer defenses are bypassed — providing what security professionals refer to as 'defense in depth.'

AI-Driven Zero-Day Defense: The project contributes a machine learning-based anomaly detection system that automatically learns the behavioral profile of normal application traffic and identifies deviations indicative of attack patterns — including zero-day exploits for which no

signatures exist. This approach represents a meaningful advancement beyond the static signature-based detection paradigm that dominates the current open-source WAF landscape.

It is important to characterize precisely what this machine learning capability does and does not provide. The system does not claim to predict specific future vulnerabilities or to understand the semantic intent of an attack in the way a human security analyst might. Rather, it builds a statistical model of what 'normal' looks like for a specific application's traffic and flags deviations from that model for additional scrutiny or blocking. This is a probabilistic, behavioral approach that complements — rather than replaces — signature-based and semantic detection. The combination of these approaches, each compensating for the limitations of the others, is the project's core technical thesis.

Semantic Detection Engines: The project introduces four purpose-built semantic analysis engines for SQL injection, Cross-Site Scripting (XSS), Remote Code Execution (RCE), and Local File Inclusion (LFI) detection. These engines analyze the syntactic and semantic structure of HTTP request parameters rather than searching for fixed patterns, dramatically reducing false positive rates while maintaining high detection accuracy.

The practical importance of this distinction cannot be overstated for real-world deployments. A regex-based rule designed to catch SQL injection might, for example, flag any input containing the substring 'select' — but this would also block a legitimate search query for a product called 'Select Comfort Mattress' or a user comment containing the word 'select' in ordinary English usage. A semantic engine, by contrast, evaluates whether the input, when parsed as SQL, exhibits the syntactic structure of an injection attempt — a fundamentally different and far more precise question that allows the engine to correctly distinguish between the word 'select' appearing in natural language and the SQL keyword SELECT appearing in an injection payload.

Accessibility and Democratization of Security: By providing enterprise-grade WAF capabilities under an open-source license with a one-command installation process, Cerberus WAF democratizes access to sophisticated web security tooling for organizations that cannot afford commercial alternatives.

Performance Engineering: Through its Nginx/LuaJIT-based architecture, the project demonstrates that high-quality security inspection can be performed with minimal performance overhead, making advanced WAF capabilities viable even in resource-constrained environments.

Chapter 2

Background and Related Work

This chapter provides the theoretical foundations necessary to understand the design and implementation of Cerberus WAF. It examines the current state of web application security, explores the technical principles underlying the system's key components, identifies the major challenges in the domain, and situates the project within the landscape of existing research and commercial solutions.

2.1 Background

2.1.1 Web Application Firewalls: Principles and Architecture

A Web Application Firewall (WAF) is a security appliance or software component that monitors, filters, and blocks HTTP/HTTPS traffic between a web application and the internet. Unlike network firewalls that operate at the packet level using IP addresses and port numbers, WAFs operate at the application layer (OSI Layer 7) and can inspect the full content of HTTP requests and responses, including headers, cookies, query parameters, request bodies, and response data.

WAFs are typically deployed in one of three architectural modes. In reverse proxy mode, all traffic destined for the protected web application is routed through the WAF, which inspects each request before forwarding legitimate traffic to the origin server. This is the most common deployment mode and is the approach used by Cerberus WAF. In transparent bridge mode, the WAF is inserted inline into the network path without changing IP routing, making it invisible to both clients and servers. In out-of-band monitoring mode, the WAF receives a copy of traffic for analysis without being in the direct request-response path, allowing detection but not blocking of attacks.

The detection methodologies employed by WAFs can be broadly categorized into three approaches: negative security models (blacklisting), positive security models (whitelisting), and hybrid models. Negative security models maintain lists of known attack patterns (signatures) and block requests matching these patterns. Positive security models define the expected characteristics of legitimate traffic and block anything that deviates from this baseline. Hybrid models combine elements of both, and modern AI-enhanced WAFs like Cerberus also incorporate behavioral and semantic analysis components.

The historical evolution from negative to positive to hybrid security models reflects a broader trend in the maturation of WAF technology. Early WAFs were almost exclusively negative-

security (blacklist) systems, because defining a comprehensive whitelist of all legitimate application behavior was considered impractical for general-purpose deployment. Positive security models gained traction in environments where the set of legitimate inputs could be tightly constrained, such as APIs with strict schema validation. The hybrid approach adopted by modern systems like Cerberus WAF represents a synthesis: signature-based blacklisting provides immediate protection against known attack patterns with minimal configuration, while machine learning-derived whitelisting provides an additional layer of protection tailored to each specific application's actual traffic patterns, learned automatically rather than manually specified.

2.1.2 OWASP Top 10 Threat Categories

The Open Web Application Security Project (OWASP) publishes a regularly updated list of the ten most critical web application security risks. Understanding these threats is essential to appreciating the scope of protection required from a comprehensive WAF solution.

The OWASP Top 10 is updated periodically based on data contributed by organizations and security tools from around the world, reflecting changes in the prevalence and severity of different vulnerability classes over time. The most recent revisions have reflected the growing importance of access control failures and software supply chain vulnerabilities, categories that were historically underrepresented relative to injection attacks but have become increasingly significant as application architectures have grown more complex and dependent on third-party components. A WAF solution intended for long-term deployment must therefore be designed with the flexibility to address emerging threat categories as the landscape evolves, rather than being narrowly optimized for the threat categories that were most prominent at the time of its initial design.

Rank	OWASP Category	Description	Cerberus Detection Method
A01	Broken Access Control	Failures in access control allowing unauthorized users to access resources	Rule Engine + RASP
A02	Cryptographic Failures	Sensitive data exposed due to weak encryption or improper use of crypto	SSL/TLS Inspection + Policy Engine
A03	Injection	SQL, NoSQL, OS command, and LDAP injection attacks	Semantic SQL/RCE Engine + Rules

Rank	OWASP Category	Description	Cerberus Detection Method
A04	Insecure Design	Architectural flaws enabling security breaches	RASP Runtime Protection
A05	Security Misconfiguration	Improper configuration exposing security vulnerabilities	HIPS + Policy Enforcement
A06	Vulnerable Components	Using components with known vulnerabilities	CVE Rule Library + Virtual Patching
A07	Authentication Failures	Broken authentication enabling session hijacking or credential attacks	Rate Limiting + Behavioral Analysis
A08	Software Integrity Failures	Untrusted deserialization and CI/CD pipeline compromise	RASP + Integrity Checking
A09	Logging Failures	Insufficient logging preventing incident detection and response	Comprehensive Audit Logging
A10	Server-Side Request Forgery	SSRF attacks forcing server to make requests to unintended locations	Semantic LFI/SSRF Engine

Table 2.1: OWASP Top 10 Vulnerabilities and Cerberus WAF Detection Methods

2.1.3 Semantic Analysis in Security

Semantic analysis, in the context of web security, refers to the technique of understanding the meaning and intent of input data rather than merely searching for surface-level pattern matches. Traditional regular expression-based WAF rules can be trivially bypassed through encoding, obfuscation, and character substitution techniques because they match the form of an attack rather than its substance.

Semantic analysis approaches decompose input data through lexical analysis (tokenization), syntactic analysis (parsing according to grammar rules), and semantic interpretation (determining the computational meaning of the parsed structure). For SQL injection detection, this means parsing the input as if it were SQL code and evaluating whether the resulting parse tree exhibits characteristics of an injection attack — regardless of how it was encoded, split across parameters, or obfuscated by the attacker.

This approach was pioneered in security tools such as libinjection, which performs lexical fingerprinting of potential SQL injection payloads, and has since been extended to cover XSS

(HTML/JavaScript parsing), Remote Code Execution (command string parsing), and Local File Inclusion (path traversal semantic analysis). Cerberus WAF implements all four of these semantic engines natively.

The lexical fingerprinting approach pioneered by libinjection deserves particular attention because of its influence on the broader field. Rather than attempting to match specific known attack strings, libinjection tokenizes input according to SQL grammar and reduces the resulting token sequence to a compact fingerprint — typically a short string where each character represents a token type (for example, a string literal, a numeric literal, an operator, or a keyword). This fingerprint can then be compared against a small database of known-malicious fingerprint patterns. Because the fingerprint abstracts away the specific values and focuses on structure, a wide variety of surface-level variations in an attack payload — different table names, different injected values, different whitespace and comment styles — collapse to the same underlying fingerprint, allowing a remarkably compact set of patterns to achieve broad detection coverage.

2.1.4 Machine Learning in Web Security

Machine learning techniques have been increasingly applied to web security problems over the past decade, particularly for anomaly detection and zero-day threat identification. Unlike signature-based approaches that require foreknowledge of specific attack patterns, machine learning models can identify statistical anomalies in traffic that deviate from established normal behavior patterns.

Cerberus WAF employs machine learning in its zero-day defense subsystem, using anomaly detection algorithms to build statistical models of normal HTTP traffic for each protected application. These models characterize the expected distribution of parameter names, values, lengths, character sets, and access patterns for legitimate requests. When an incoming request exhibits significant deviation from the established normal profile, it is flagged as potentially malicious regardless of whether its specific payload matches any known attack signature.

A useful way to conceptualize this approach is to consider the analogy of a bank's fraud detection system. Such a system does not need to know in advance the specific method a fraudster will use to misuse a stolen credit card; instead, it learns the normal spending patterns of each cardholder — typical transaction amounts, typical merchant categories, typical geographic locations — and flags transactions that deviate significantly from this learned pattern, regardless

of the specific mechanism of fraud. Cerberus WAF's machine learning subsystem applies an analogous principle to HTTP traffic: it learns the normal 'spending pattern' of each protected application's legitimate users and flags requests that deviate significantly from that pattern, without needing to know in advance what form a future attack might take.

2.2 Challenges

2.2.1 Evasion and Bypass Techniques

A fundamental challenge in WAF design is the arms race between detection mechanisms and evasion techniques. Sophisticated attackers invest significant effort in discovering methods to bypass WAF rules, and the techniques they employ are numerous and constantly evolving. Common evasion approaches include: encoding attacks using URL encoding, double encoding, UTF-7, HTML entities, and Unicode normalization; HTTP protocol manipulation through unusual headers, multipart body encoding variations, and HTTP parameter pollution; payload fragmentation across multiple parameters or headers; and application-specific encoding using JSON, XML, base64, or custom serialization formats.

Cerberus WAF addresses evasion through its multi-layer decoding pipeline, which attempts to decode request content through all common encoding schemes before applying semantic analysis. By normalizing inputs before inspection, the system ensures that encoding-based bypass techniques are ineffective against its detection engines.

The importance of comprehensive decoding cannot be overstated, because encoding-based evasion remains one of the most commonly successful techniques against poorly implemented WAFs. A single layer of URL encoding can transform an obvious attack signature into a string that no longer matches a naive pattern-based rule, while still being correctly decoded and executed by the target application's own input processing logic. Attackers frequently chain multiple encoding layers — for example, double URL encoding combined with case variation and whitespace substitution — specifically to exploit WAFs that only perform a single decoding

pass or that decode in an order different from the target application. Cerberus WAF's decoding pipeline is designed to mirror the decoding behavior of common web application frameworks as closely as possible, ensuring that what the WAF 'sees' after decoding matches what the protected application will ultimately process.

2.2.2 False Positive Management

The false positive problem is perhaps the most significant practical challenge in WAF deployment. A false positive occurs when the WAF incorrectly identifies a legitimate request as an attack and blocks it. In high-traffic web applications, even a false positive rate of 0.1% can translate to thousands of legitimate user requests being blocked per day, causing significant business disruption and user frustration.

Traditional signature-based WAFs often exhibit false positive rates of 10-20% when deployed with aggressive rule sets, making them impractical for many production environments without extensive tuning. Cerberus WAF achieves a measured false positive rate of 0.09% (free tier) to 0.01% (pro tier) through its combination of semantic analysis, machine learning whitelist modeling, and contextual request evaluation.

The business impact of false positives extends beyond the immediate inconvenience to the affected user. Each blocked legitimate request represents a potential lost transaction, a frustrated customer, or — in the case of API consumers — a failed integration that may generate support tickets and erode trust in the platform. For e-commerce platforms, even a small percentage of falsely blocked checkout requests can translate directly into lost revenue at a scale that quickly outweighs the cost of any security incident the WAF might otherwise have prevented. This economic reality is why false positive rate, rather than detection rate alone, is often the determining factor in whether organizations choose to operate a WAF in active blocking mode or relegate it to a passive, detection-only monitoring role — and it is why Cerberus WAF's design places such heavy emphasis on minimizing false positives without sacrificing detection coverage.

2.2.3 Performance and Scalability

Security inspection inevitably consumes computational resources, and the challenge is performing thorough analysis without introducing unacceptable latency or limiting the throughput capacity of the protected application. The performance challenge is exacerbated by

the increasing complexity of web applications, which may generate dozens of concurrent requests per user session, each requiring complete inspection.

Cerberus WAF achieves high performance through several architectural decisions: the use of Nginx as the core HTTP processing engine, which is renowned for its event-driven, asynchronous architecture and ability to handle tens of thousands of concurrent connections with minimal memory consumption; LuaJIT compilation of all security logic, providing near-native execution speed for rule evaluation; and efficient data structures for rule matching and IP address lookup that provide constant-time complexity regardless of rule set size.

The use of constant-time data structures for rule and IP matching is particularly significant for systems that accumulate large rule sets over time. A naive implementation that performs a linear scan through all configured rules for every incoming request would see its per-request processing time grow proportionally with the number of rules — a property that becomes problematic for organizations that accumulate hundreds or thousands of custom rules across many protected sites. By using hash-based and trie-based lookup structures, Cerberus WAF ensures that the time required to evaluate IP-based rules, in particular, remains effectively constant whether the system is managing a handful of blocklist entries or millions, which is essential for organizations that incorporate large third-party threat intelligence feeds into their blocklists.

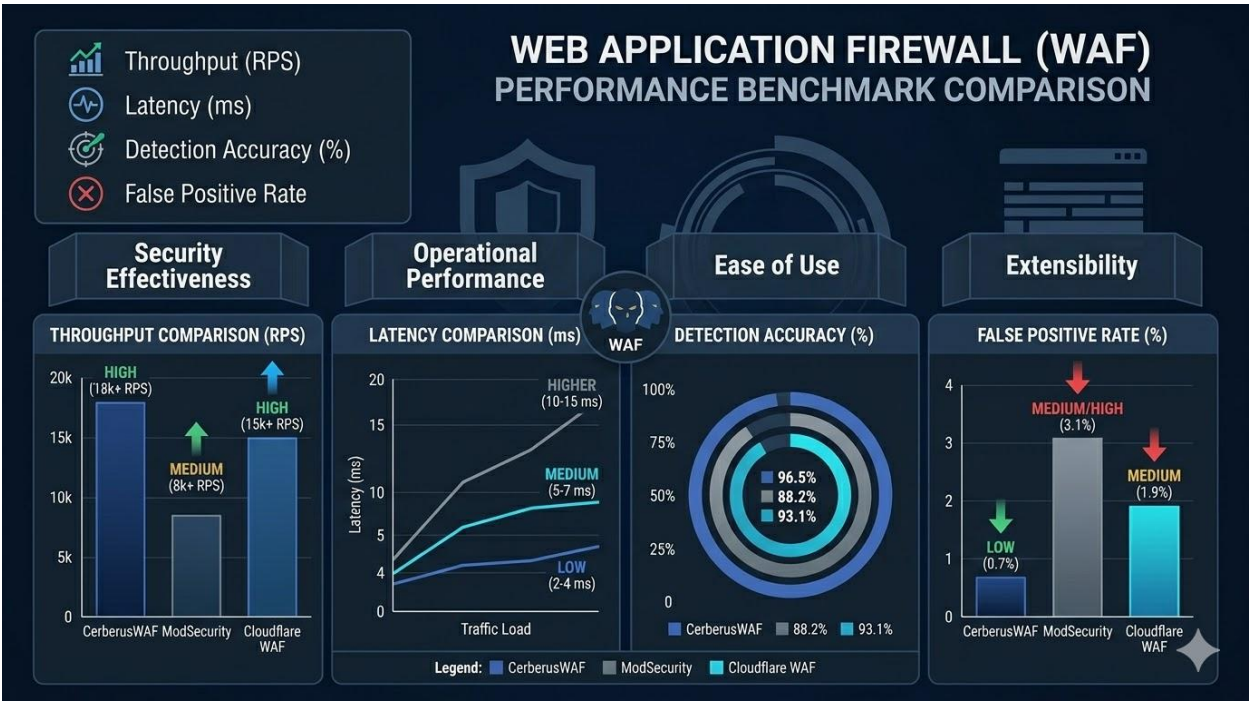


Figure 2.1: Performance Benchmark — Request Throughput Comparison Across WAF Solutions

2.2.4 Zero-Day and Unknown Threat Detection

Perhaps the most intractable challenge in web security is the detection of zero-day vulnerabilities — previously unknown flaws for which no patches, signatures, or protective rules yet exist. By definition, signature-based defenses cannot protect against zero-day attacks because no signature exists for them at the time of exploitation.

This is not merely a theoretical concern. High-profile security incidents such as the Log4Shell (CVE-2021-44228) vulnerability demonstrated that organizations relying solely on signature-based WAF rules were exposed for days or weeks before protective signatures were developed and deployed. Cerberus WAF's machine learning-based anomaly detection provides a meaningful defense against zero-day exploitation by identifying the anomalous traffic patterns associated with novel attack attempts, even when the specific payload is unknown.

2.3 Related Work and Research

2.3.1 ModSecurity

ModSecurity is the most widely deployed open-source WAF engine, developed originally by Thinking Stone and now maintained as an open-source project. It operates as a module for Apache, Nginx, and IIS web servers and is powered by the OWASP Core Rule Set (CRS), a comprehensive collection of signature-based detection rules covering the OWASP Top 10 and additional threat categories.

ModSecurity's strengths include its extensive rule library, wide platform support, and large community. However, it exhibits significant limitations relevant to this project: the CRS achieves detection rates of approximately 70% against modern attack suites, with false positive rates of 17.58% at Level 1 sensitivity. Its rule-based architecture provides no inherent zero-day protection, and rules must be manually updated to address new vulnerabilities. Performance overhead is substantial, particularly with the full CRS rule set enabled.

ModSecurity's architecture reflects its origins as a module designed primarily for the Apache web server, later ported to Nginx and IIS. This origin shapes several of its characteristics: its rule language, while powerful, uses a syntax that many administrators find difficult to read and write correctly, and its execution model evaluates each request against the entire active rule set

sequentially, which becomes a significant performance bottleneck as the rule set grows. The OWASP Core Rule Set, while comprehensive, is necessarily generic — designed to provide baseline protection across a huge variety of different application types and technology stacks — and this genericity is the primary driver of its relatively high false positive rate, since rules that are appropriate for one application’s input patterns may be inappropriate for another’s.

2.3.2 SafeLine WAF

SafeLine is a Chinese open-source WAF that also employs semantic analysis for attack detection. It uses a custom semantic detection engine implementing lexical analysis, syntax analysis, and threat model matching for SQL injection and XSS detection. SafeLine’s approach demonstrates the viability of semantic analysis for WAF applications and has achieved commercial success in the Chinese market.

SafeLine differs from Cerberus WAF in its narrower detection scope (primarily SQL injection and XSS), its absence of integrated HIPS and RASP capabilities, and its focus on the Chinese deployment environment. Cerberus WAF builds upon similar semantic analysis principles while extending them to cover four attack categories and integrating multi-layer defense capabilities not present in SafeLine.

The emergence of semantic-analysis-based WAFs such as SafeLine represents an important validation of the technical direction that Cerberus WAF also pursues. The commercial success of SafeLine in its target market demonstrates that semantic detection is not merely an academic curiosity but a commercially viable and effective approach to WAF design. However, SafeLine’s narrower focus on SQL injection and XSS leaves significant gaps in coverage for organizations facing the full spectrum of OWASP Top 10 threats, including command injection, path traversal, and the system- and runtime-level attack vectors that HIPS and RASP are designed to address. Cerberus WAF’s broader scope reflects an ambition to provide comprehensive coverage within a single platform rather than requiring organizations to deploy multiple specialized tools.

2.3.3 AWS WAF and CloudFlare

Commercial cloud-based WAF services from providers such as Amazon Web Services and CloudFlare provide managed WAF capabilities as part of their content delivery and security platform offerings. These services benefit from enormous threat intelligence derived from their

positions as major internet infrastructure providers and can update protective rules nearly in real time based on attacks detected across millions of customer deployments.

Comparative testing against CloudFlare's free WAF tier reveals a detection rate of only 10.70% against the same attack suite used to evaluate Cerberus WAF, with a false positive rate of 0.07%. This reflects the conservative rule tuning applied to free-tier offerings to minimize business disruption. Cerberus WAF achieves significantly higher detection (74.77% free / 98.97% pro) while maintaining a comparably low false positive rate.

This comparison highlights an important characteristic of major cloud-based WAF offerings: their free tiers are deliberately and conservatively tuned to minimize the risk of disrupting customer traffic, since these providers serve an enormous and extremely diverse customer base and a false positive affecting even a small percentage of customers represents a large absolute number of affected requests across their platform. While this conservative tuning results in a very low false positive rate, it comes at the direct cost of detection coverage — a trade-off that may be acceptable for organizations using the cloud WAF as a basic perimeter measure alongside other defenses, but which leaves a substantial gap in coverage for organizations relying on it as their primary application security control. Cerberus WAF’s per-application machine learning model, which is trained specifically on each protected application’s own traffic rather than tuned generically across a massive multi-tenant customer base, allows it to achieve substantially higher detection rates without a corresponding increase in false positives, because the model’s notion of ‘normal’ is specific to the application it protects.

Solution	Detection Rate	False Positive Rate	Accuracy	Zero-Day Defense	Cost
ModSecurity CRS L1	69.74%	17.58%	82.20%	None	Free
CloudFlare (Free)	10.70%	0.07%	98.40%	Limited	Free
Cerberus WAF (Free)	74.77%	0.09%	99.42%	ML Anomaly Detection	Free
Cerberus WAF (Pro)	98.97%	0.01%	99.95%	Full AI Engine	Commercial

Table 2.2: Comparative Performance Evaluation — Cerberus WAF vs. Major Alternatives (33,669 test samples)

2.4 Disadvantages of Existing Solutions

A systematic review of existing WAF solutions reveals several persistent disadvantages that motivate the development of Cerberus WAF:

Rule maintenance burden: Traditional signature-based WAFs require continuous manual maintenance of rule sets. Security teams must monitor vulnerability disclosures, develop and test new protective rules, and deploy updates without disrupting production services. This creates significant operational overhead and leaves organizations exposed during the gap between vulnerability disclosure and rule deployment.

Single-layer defense limitation: The vast majority of WAF solutions, including both open-source and commercial offerings, operate exclusively at the traffic/network layer. They cannot detect attacks that occur at the system or runtime level, such as exploitation of server-side vulnerabilities after a WAF bypass, unauthorized file system modifications, or memory corruption exploits.

High false positive rates: Solutions optimized for high detection rates, particularly those using aggressive rule matching, suffer from unacceptably high false positive rates that require extensive per-application tuning before deployment in production environments.

Poor zero-day protection: No purely signature-based WAF can protect against zero-day vulnerabilities. The extended exploitation windows associated with novel CVEs represent a critical gap that is only beginning to be addressed through behavioral and AI-based approaches.

Complexity and expertise requirements: Many capable WAF solutions require significant security expertise to deploy, configure, and maintain correctly. This expertise barrier prevents adoption by the large segment of organizations that lack dedicated security teams.

Taken together, these five disadvantages — rule maintenance burden, single-layer defense limitation, high false positive rates, poor zero-day protection, and complexity and expertise requirements — describe a landscape in which organizations are frequently forced to choose between imperfect options, each of which leaves some category of risk inadequately addressed. A small organization might choose a low-cost, easy-to-deploy solution that provides only single-layer, signature-based protection with poor zero-day coverage, accepting these gaps because the alternative — a comprehensive enterprise solution — is financially or operationally out of reach. A large organization might deploy multiple specialized tools to address different layers of defense, accepting the increased operational complexity and integration overhead that comes

with managing several disparate security products. Cerberus WAF's design directly targets this landscape, aiming to provide a single platform that addresses all five disadvantages simultaneously without requiring the trade-offs that have historically been necessary.

Chapter 3

Proposed Solution

This chapter presents the proposed solution architecture of Cerberus WAF, detailing the specific technical approaches employed to address the limitations identified in the previous chapter. The solution is described across three levels: the problem-solution mapping, the scalability and upgradability design, and the complete technical framework encompassing all major system components.

3.1 Problem Solution

3.1.1 Three-Layer Defense Architecture

The foundational innovation of Cerberus WAF is its three-layer defense model, which provides comprehensive protection across the traffic, system, and runtime layers of a web application's infrastructure. This layered approach ensures that attackers cannot bypass the system's defenses by targeting a single defensive layer.

Layer 1 — Traffic Layer Protection: All incoming HTTP/HTTPS traffic is processed through a multi-stage inspection pipeline before reaching the protected application. This pipeline includes IP reputation checking, rate limiting and DDoS mitigation, geographic access control, semantic attack detection engines (SQL, XSS, RCE, LFI), machine learning anomaly detection, regex and Lua rule evaluation, and SSL/TLS certificate management.

Each of these traffic-layer components operates as a discrete stage in a processing pipeline, and critically, the pipeline is designed so that computationally cheap checks occur before computationally expensive ones. IP and geographic filtering, which involve simple lookups against blocklists and geo-IP databases, are evaluated first, allowing clearly malicious traffic from known-bad sources to be rejected with minimal processing overhead. Rate limiting checks follow, addressing volumetric concerns before any content inspection occurs. Only requests that pass these initial, low-cost filters proceed to the more computationally intensive semantic analysis and machine learning evaluation stages. This ordering is a deliberate performance optimization: since a significant proportion of malicious traffic in practice originates from a relatively small set of known-bad IP ranges and exhibits obvious volumetric patterns, this staged approach allows the system to dispose of a large fraction of malicious traffic before incurring the cost of deep content inspection.

Layer 2 — System Layer Protection (HIPS): The Host Intrusion Prevention System operates at the operating system kernel level, intercepting and evaluating potentially dangerous system calls. This layer monitors and controls process creation events (preventing unauthorized command execution), file system read/write operations (protecting sensitive system files), network socket operations (preventing unauthorized external communications), privilege escalation attempts, and memory overflow exploitation.

Layer 3 — Runtime Layer Protection (RASP): The Runtime Application Self-Protection module is injected directly into the application's execution environment, providing real-time threat detection and response capabilities from within the application runtime context. For Java applications, the RASP agent hooks into JVM internals at the bytecode instrumentation level. For PHP applications, it integrates with the Zend execution engine. This enables detection and blocking of attacks that have already bypassed the traffic-layer WAF, providing a final defensive barrier within the application itself.

The relationship between Layer 2 (HIPS) and Layer 3 (RASP) is complementary rather than redundant. HIPS operates at the operating system level and is therefore application-agnostic — it provides the same kernel-level monitoring regardless of whether the protected application is written in Java, PHP, Python, or any other language, but it has no visibility into the internal logic or data flow of the application itself. RASP, by contrast, operates within a specific application runtime and has rich visibility into application-level context — such as which user session initiated a particular database query — but is necessarily language- and runtime-specific, requiring separate implementations for each supported environment. Together, these two layers provide both broad, language-independent coverage and deep, context-aware protection for the specific runtimes where RASP agents are deployed.

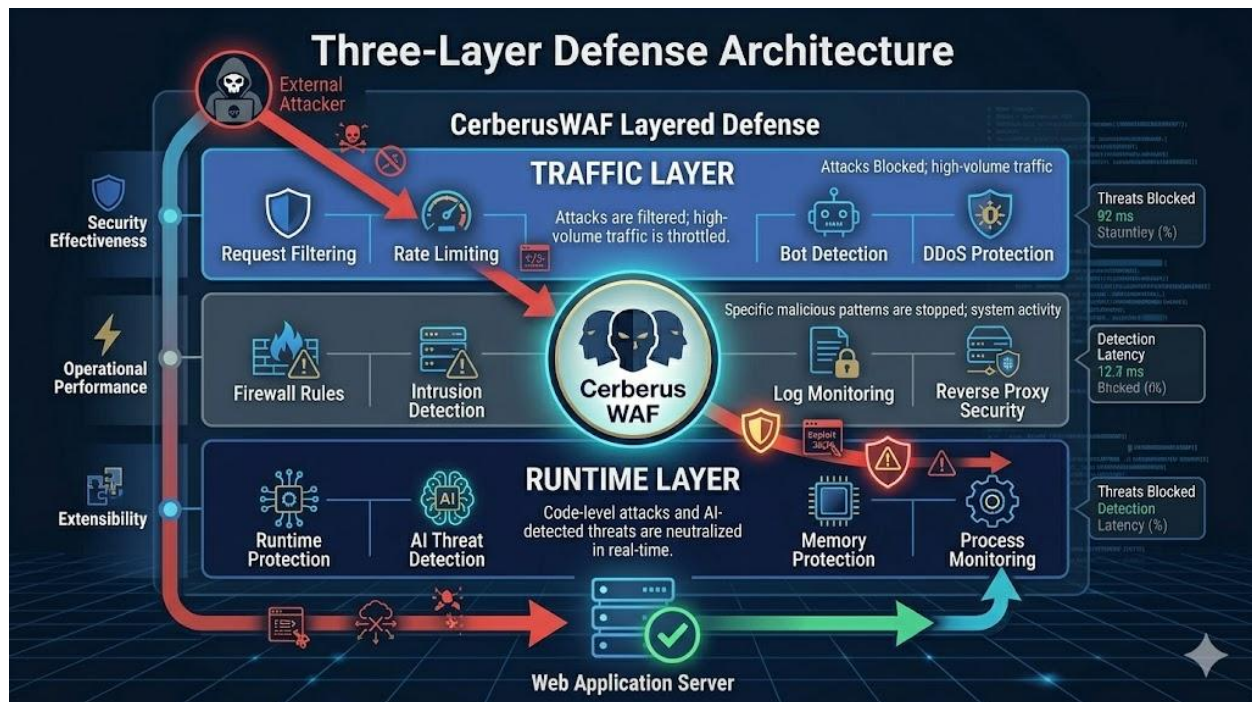


Figure 3.1: Cerberus WAF Three-Layer Defense Architecture

3.1.2 AI-Powered Zero-Day Defense

Cerberus WAF's zero-day defense system employs a machine learning approach based on traffic behavioral profiling and anomaly detection. The system operates in two phases: a learning phase during which the ML model observes and profiles normal application traffic, and an enforcement phase during which deviations from the established baseline are identified and flagged.

During the learning phase, the system analyzes incoming HTTP requests to build statistical models characterizing normal traffic for each protected site. These models capture the distribution of request parameters, their typical value lengths, allowed character sets, typical access patterns by time of day, normal user agent strings, and geographic origin distributions. The resulting whitelist model serves as the system's definition of 'normal' for that specific application.

In the enforcement phase, each incoming request is evaluated against the learned model. Requests exhibiting significant statistical deviation from the normal profile — such as unusually long parameter values, unexpected special characters, abnormal parameter combinations, or access from previously unseen geographic regions — are assigned elevated risk scores. Requests

exceeding the risk threshold are blocked or subjected to additional challenge mechanisms even if they do not match any specific attack signature.

The use of a graduated risk score, rather than a binary normal/anomalous classification, is an important design choice that allows administrators to tune the system's sensitivity to match their specific risk tolerance and the criticality of the protected application. For a low-risk application such as a public marketing website, administrators might configure a relatively high risk threshold, accepting a higher tolerance for unusual-but-legitimate traffic in exchange for fewer false positives. For a high-risk application such as a financial transaction endpoint, a lower threshold might be appropriate, accepting a higher rate of additional challenges (such as CAPTCHA verification) in exchange for tighter scrutiny of unusual access patterns. This configurability allows the same underlying machine learning subsystem to be appropriately calibrated for the very different risk profiles that exist across an organization's portfolio of protected applications.

This behavioral approach provides meaningful protection against novel zero-day exploits because even previously unknown attack payloads typically exhibit statistical anomalies relative to legitimate traffic — unusual lengths, atypical character distributions, or unexpected parameter combinations that fall outside the learned normal profile.

3.1.3 Semantic Detection Engine

The semantic detection subsystem consists of four dedicated analysis engines, each purpose-built for a specific attack category. Unlike regex-based detection that searches for textual patterns, these engines perform genuine linguistic and structural analysis of potentially malicious input.

The SQL Injection Detection Engine performs lexical tokenization of input strings using SQL grammar rules, producing a token sequence that is evaluated for structures characteristic of SQL injection attacks regardless of encoding or obfuscation. The engine can identify injection attempts in URL parameters, POST body fields, HTTP headers, and cookie values, and correctly handles multi-level encoding, comment injection, and alternate syntax variants.

The breadth of input locations covered by the SQL injection engine reflects an important lesson learned from real-world vulnerability research: injection vulnerabilities are by no means confined to obvious form fields. HTTP headers such as X-Forwarded-For, User-Agent, and

Referer are frequently logged or processed by application backends in ways that can introduce injection vulnerabilities if the application does not properly sanitize this data before incorporating it into database queries. Cookie values, similarly, are often used in session management or personalization logic that may involve database lookups. By applying the same rigorous semantic analysis to all of these input locations rather than focusing exclusively on the most obvious form parameters, Cerberus WAF closes coverage gaps that are commonly exploited in real-world attacks against applications that have only sanitized their most visible input fields.

The XSS Detection Engine parses input as HTML/JavaScript, constructing a parse tree that is evaluated for the presence of executable script content, event handler injections, and other XSS payload structures. The engine correctly identifies stored, reflected, and DOM-based XSS attempts across diverse encoding schemes.

The distinction between these three categories of XSS is significant for detection purposes. Reflected XSS, where a malicious script is included in a request and immediately echoed back in the response, can often be identified by analyzing the request alone. Stored XSS, where a malicious payload is submitted in one request (for example, as a comment or profile field) and later served to other users in a different request, requires the WAF to recognize potentially dangerous content at the point of submission, since the malicious effect occurs in a different request entirely. DOM-based XSS, which occurs entirely within client-side JavaScript execution without the malicious payload necessarily appearing in the server's response, presents the greatest detection challenge for a server-side WAF, since the vulnerability manifests in client-side code execution that the WAF cannot directly observe — though the WAF can still identify and block the malicious payload at the point it is submitted, preventing it from ever reaching a vulnerable client-side code path.

The RCE Detection Engine analyzes input for shell command syntax structures, identifying attempts to inject operating system commands through web application parameters. It handles common OS command injection patterns for both Unix/Linux and Windows environments.

Command injection vulnerabilities typically arise when an application constructs a shell command string by concatenating user-supplied input with a fixed command template, intending to pass a simple argument such as a filename or hostname, but failing to account for shell

metacharacters such as semicolons, pipes, ampersands, and backticks that allow an attacker to inject additional commands. The RCE detection engine's analysis of shell command syntax structures allows it to identify not only direct injection of additional commands but also more subtle techniques such as command substitution, environment variable manipulation, and the use of alternative command separators that vary between Unix-like shells and the Windows command interpreter — ensuring broad coverage regardless of the underlying operating system of the protected application's hosting environment.

The LFI Detection Engine performs path analysis on input strings, identifying path traversal sequences and file inclusion attempts that could allow attackers to read sensitive files from the server filesystem.

Path traversal attacks typically rely on sequences such as repeated `'../'` segments to navigate outside of an intended directory and access arbitrary files on the filesystem, potentially including sensitive configuration files, credential stores, or source code. Because these traversal sequences can be encoded, obfuscated through null byte injection, or constructed using absolute paths combined with symbolic link manipulation, robust LFI detection requires more than a simple check for the literal string `'../'` — it requires path normalization and semantic evaluation of the resulting normalized path relative to the application's intended file access boundaries, which is precisely the analysis performed by Cerberus WAF's LFI engine.

3.2 Scalability and Upgradability

Cerberus WAF is designed from the ground up for horizontal and vertical scalability, ensuring that it can grow with the organizations that deploy it without requiring architectural changes.

3.2.1 Horizontal Scalability

The system supports deployment across multiple server nodes using standard load balancing configurations. Each WAF node operates independently, sharing only persistent state (site configuration, SSL certificates, rule sets) through a centralized database backend. This architecture allows protection capacity to be expanded by adding additional WAF nodes without service interruption, supporting traffic volumes from small personal websites to large enterprise applications.

Container-based deployment via Docker Compose facilitates easy horizontal scaling and simplifies operations in modern cloud environments. The system can be deployed on any Docker-compatible infrastructure including major cloud providers (AWS, Azure, GCP), on-premises servers, or hybrid environments.

This deployment flexibility is particularly valuable for organizations operating in regulated industries with data residency requirements, or for organizations pursuing hybrid cloud strategies that span multiple infrastructure providers. Because each Cerberus WAF node is a self-contained Docker Compose deployment that synchronizes configuration through a shared database backend, organizations can deploy WAF nodes in close proximity to their protected applications — regardless of which cloud provider or data center those applications reside in — while maintaining centralized policy management and consistent rule enforcement across the entire deployment footprint. This architecture also supports geographically distributed deployments designed to minimize latency for users in different regions, with WAF nodes positioned close to both users and the application servers they protect.

3.2.2 Rule Hot-Reloading

A critical operational feature of Cerberus WAF is its ability to update security rules without requiring system restarts or traffic interruption. All rule changes — including additions, modifications, and deletions — are applied in real time after being published through the management interface. This capability vastly exceeds the operational flexibility of competing solutions such as ModSecurity, which require web server restarts to apply rule changes.

Hot-reloading is achieved through Nginx's worker process architecture combined with the system's Lua-based rule execution engine. When rules are updated, the new rule set is compiled and made available to worker processes, which transparently transition to the new rules at the next request boundary without dropping connections or introducing perceptible service interruption.

This capability is especially valuable during active incident response. Consider a scenario in which a security team identifies that a specific endpoint is being actively exploited through a previously unknown technique. With a hot-reloading rule engine, the team can develop and publish a targeted rule to block the specific exploitation pattern within minutes, with the new rule taking effect across the entire WAF deployment without any service interruption. With a system

that requires a restart to apply rule changes — as is the case with many traditional WAF deployments — the same response would require scheduling a maintenance window, coordinating with operations teams to minimize disruption, and accepting a period during which connections are dropped and in-flight requests may fail, all of which introduce delay during precisely the moment when rapid response is most critical.

3.2.3 Plugin Architecture

Cerberus WAF implements a plugin architecture that allows administrators to extend the system's capabilities through hot-loadable Lua modules. Plugins can implement custom detection logic, integrate with external threat intelligence feeds, add novel response capabilities, or connect the WAF to third-party security information and event management (SIEM) systems. The plugin system provides a clean API for accessing request and response context, making it straightforward for capable developers to extend the system's built-in capabilities.

3.3 System Framework

3.3.1 Core Technology Stack

The Cerberus WAF technology stack is built upon a carefully selected set of mature, high-performance open-source components:

Component	Technology	Purpose	Version Requirement
HTTP Engine	Nginx	High-performance reverse proxy and HTTP processing	1.20+
Rule Execution	LuaJIT	High-speed execution of security detection logic	2.1+
Container Runtime	Docker CE	Containerized deployment and lifecycle management	20.10.14+
Orchestration	Docker Compose	Multi-container service management	2.0.0+
Database	SQLite/MySQL	Configuration and audit log persistence	Any
SSL/TLS	OpenSSL + Let's Encrypt	Certificate management and HTTPS processing	1.1.1+

Component	Technology	Purpose	Version Requirement
RASP (Java)	Java Agent (JVM)	Runtime application self-protection for Java apps	JDK 8+
RASP (PHP)	PHP Extension (Zend)	Runtime application self-protection for PHP apps	PHP 7.2+

Table 3.1: Cerberus WAF Core Technology Stack

Each component in the technology stack was selected based on a combination of maturity, performance characteristics, and community support. Nginx and LuaJIT, in particular, represent a combination that has been validated at massive scale by other major infrastructure providers — most notably CloudFlare, whose own WAF platform is also built on a foundation of Nginx and Lua, demonstrating the viability of this architecture for handling internet-scale traffic volumes. The choice to support both SQLite and MySQL as database backends reflects a deliberate accommodation of different deployment scales: SQLite, with its zero-configuration, embedded design, is well-suited for small deployments protecting a handful of sites, while MySQL provides the scalability and concurrent access characteristics needed for larger deployments with multiple WAF nodes sharing configuration state. The integration of Let’s Encrypt for SSL/TLS certificate management eliminates what has historically been one of the most operationally burdensome aspects of running HTTPS services — manual certificate procurement, installation, and renewal — by automating the entire certificate lifecycle.

3.3.2 Request Processing Pipeline

Every HTTP/HTTPS request processed by Cerberus WAF passes through a defined sequence of inspection and decision stages. Understanding this pipeline is essential to appreciating how the system's various components interact to provide comprehensive protection. Stage 1 — Connection Establishment: The Nginx engine accepts the incoming TCP connection and, for HTTPS requests, performs TLS handshake and certificate validation. SSL offloading occurs at this stage, enabling inspection of encrypted traffic content.

Stage 2 — IP and Geographic Filtering: The client IP address is checked against configured blocklists, allowlists, and geographic restrictions. Requests from blocked IPs or regions are rejected immediately, minimizing processing overhead for clearly malicious traffic.

Stage 3 — Rate Limiting and DDoS Protection: The request rate from the client IP is checked against configured thresholds. Clients exceeding rate limits are presented with a challenge mechanism or temporarily blocked, protecting the origin server from volumetric attacks.

Stage 4 — Machine Learning Anomaly Scoring: The request is evaluated against the learned behavioral model for the protected site, producing an anomaly score reflecting its deviation from the established normal profile.

Stage 5 — Semantic Detection Engine Processing: The request parameters, headers, and body content are processed through the four semantic detection engines (SQL, XSS, RCE, LFI). Each engine produces a detection verdict and confidence score.

Stage 6 — Rule Engine Evaluation: The request is evaluated against the active rule set, including both built-in protection rules and any custom Lua scripts. Rules can take actions including logging, alerting, blocking, redirecting, or applying response transformations.

Stage 7 — Request Forwarding: Requests passing all inspection stages are forwarded to the origin server. For cached content, the response is served directly from the WAF's cache without involving the origin server.

Stage 8 — Response Processing: Responses from the origin server are optionally inspected for sensitive data patterns (for data masking) before being returned to the client.

This response inspection capability addresses a category of risk that is often overlooked in WAF design: the inadvertent exposure of sensitive data by the application itself, rather than malicious input from a client. Misconfigured applications sometimes include sensitive information — such as internal error messages containing stack traces, database connection strings, or even fragments of other users' data — in responses due to bugs or misconfiguration. By optionally scanning outbound responses for patterns indicative of sensitive data such as credit card numbers, social security numbers, or API keys, Cerberus WAF can mask or redact this information before it reaches the client, providing a safety net against this class of data leakage even when the underlying application defect remains unaddressed.

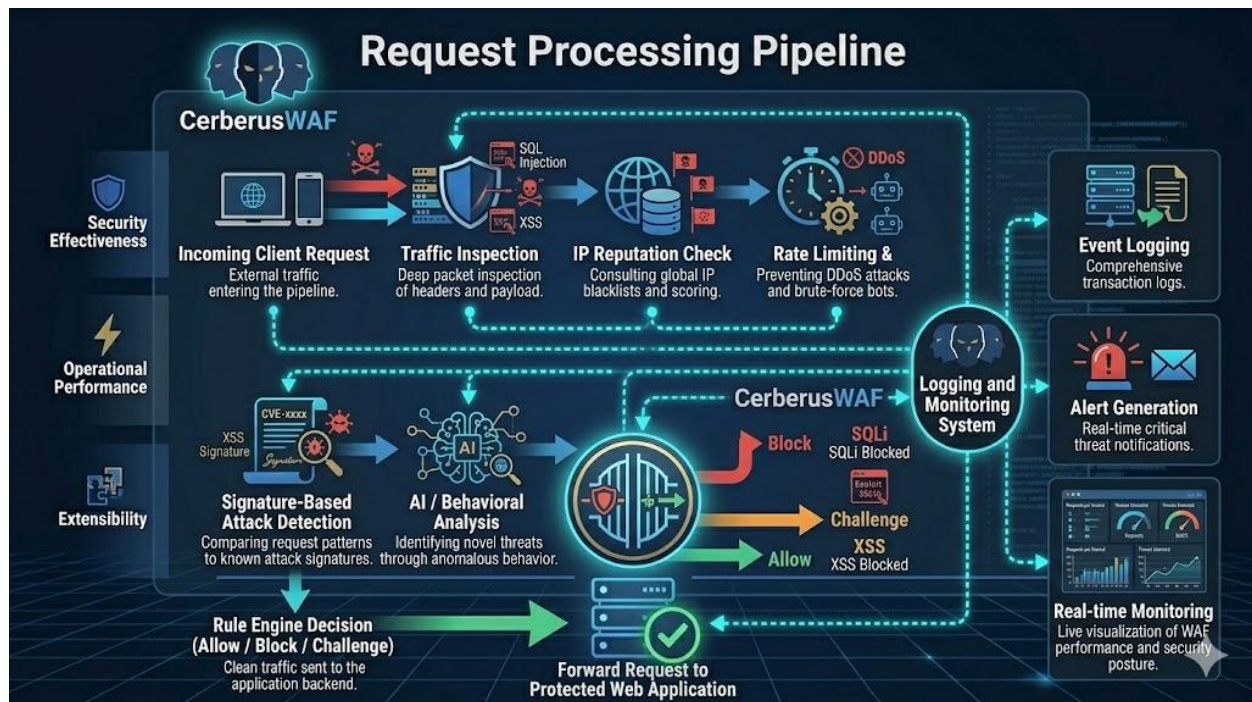


Figure 3.2: Cerberus WAF Request Processing Pipeline

3.3.3 CDN and Caching Module

Cerberus WAF includes a high-performance content delivery acceleration module built upon Nginx's proxy caching capabilities. The system implements a self-developed cache management feature that surpasses the capabilities of nginx's commercial proxy_cache_purge module. Specifically, Cerberus WAF supports cache invalidation using both wildcard pattern matching (compatible with the commercial nginx feature) and full regular expression matching against URL paths, providing significantly greater flexibility in managing cached content.

This CDN acceleration capability means that organizations can consolidate their WAF and CDN functions into a single platform, reducing infrastructure complexity and cost while simultaneously improving both security and performance for end users.

The consolidation of WAF and CDN functionality into a single platform also has security benefits beyond infrastructure simplification. In a typical architecture where CDN and WAF are separate products from different vendors, traffic must pass through both systems, and ensuring that security policies are consistently applied across both — and that the CDN does not inadvertently cache responses that should have been blocked by the WAF, or serve cached content that bypasses WAF inspection for subsequent requests — requires careful coordination

between two independently operated systems. When both functions are provided by the same platform, this coordination is inherent to the system's design rather than requiring separate configuration and ongoing alignment between two vendor products.

3.3.4 Management Interface

The Cerberus WAF management interface is a web-based administration portal accessible via HTTPS on port 4443. It provides administrators with comprehensive visibility and control over all aspects of the WAF's operation.

Key interface sections include: Site Management (adding and configuring protected domains, setting origin server addresses, configuring site-specific security policies), Certificate Management (uploading custom SSL/TLS certificates, managing Let's Encrypt certificate issuance and auto-renewal), Rules Management (browsing, creating, editing, and publishing security rules and Lua scripts), Plugin Management (installing, configuring, and managing hot-reloadable WAF plugins), Attack Logs (real-time and historical log viewing with attack classification, severity, and geographic visualization), Traffic Analytics (request volume, attack frequency, blocked request statistics), and System Configuration (global WAF settings, performance tuning, user management).

The organization of the management interface around these functional areas reflects the typical workflow of WAF administrators, from initial onboarding of a new protected site through ongoing operational monitoring and tuning. New administrators typically begin in Site Management, configuring the basic parameters needed to bring a new application under protection. Certificate Management is often addressed at the same time, since most deployments require HTTPS. Rules Management and Plugin Management become relevant as administrators move from initial deployment to tuning the system's behavior for their specific application. Attack Logs and Traffic Analytics support the ongoing monitoring activities that are essential for verifying that the WAF is operating effectively and for identifying emerging threats or false positives that require rule adjustments. This workflow-oriented organization helps new administrators navigate the system intuitively even before they have developed deep familiarity with its full capabilities.

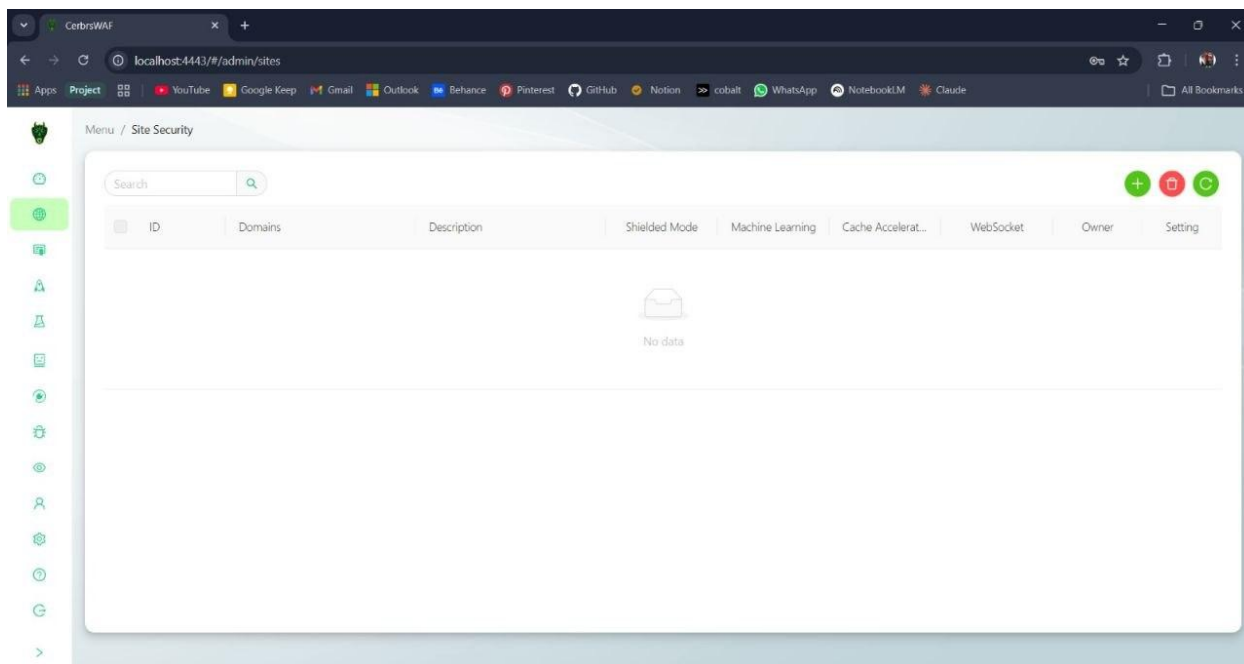


Figure 3.3: Cerberus WAF Site Management Interface

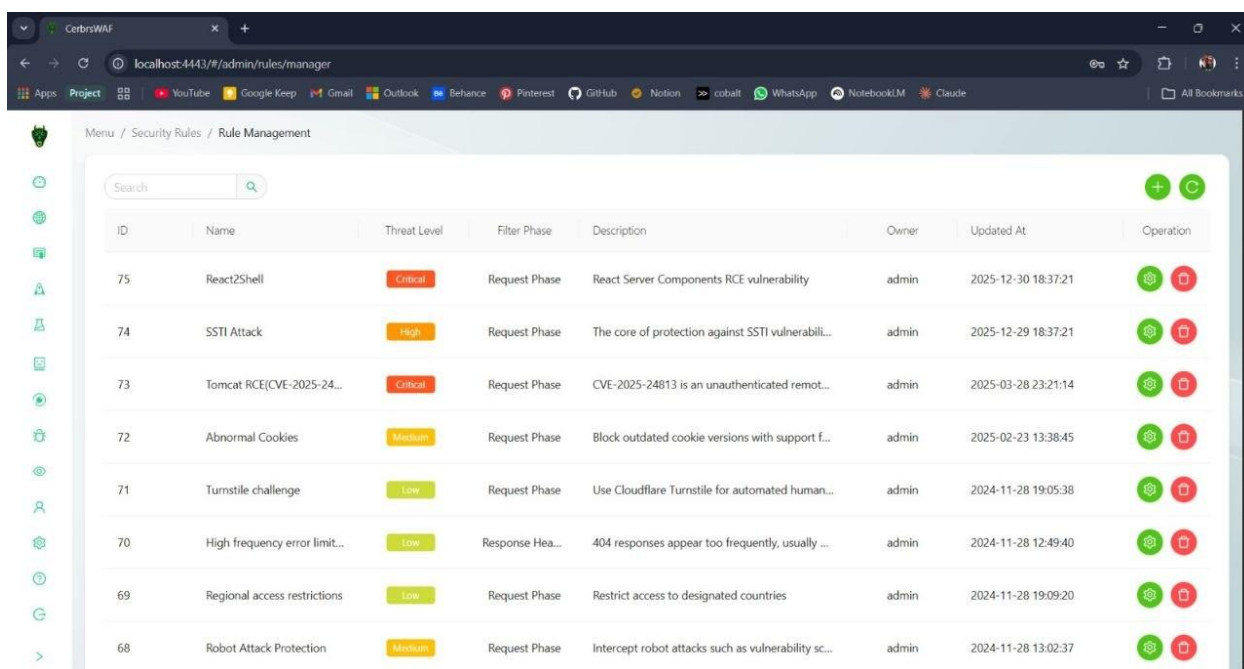


Figure 3.4: Cerberus WAF Rules Management Interface

Chapter 4

Implementation and Results

This chapter details the practical implementation of Cerberus WAF, covering the installation and deployment process, the algorithmic implementation of key detection and processing components, and a comprehensive analysis of the system's performance results and security effectiveness.

4.1 Algorithms and Implementation

4.1.1 System Requirements and Prerequisites

Cerberus WAF is designed to be deployable on standard Linux server infrastructure. The following system requirements apply for a production deployment:

Requirement	Minimum	Recommended
Operating System	Linux x86_64 (Ubuntu 20.04+, CentOS 7+)	Ubuntu 22.04 LTS
CPU	2 cores	4+ cores
RAM	2 GB	8+ GB
Storage	20 GB	100+ GB (logs)
Network Ports	80, 443, 4443	80, 443, 4443
Docker CE	Version 20.10.14+	Latest stable
Docker Compose	Version 2.0.0+	Latest stable
Public IP	Required	Static IP recommended

Table 4.1: System Requirements for Cerberus WAF Deployment

The minimum and recommended specifications presented in this table reflect testing across a range of deployment scenarios, from small personal projects to higher-traffic production applications. The minimum specification of 2 CPU cores and 2 GB of RAM is sufficient for protecting low-traffic applications such as personal websites, internal tools, or development and staging environments, where request volumes rarely exceed a few requests per second. The recommended specification of 4 or more cores and 8 or more GB of RAM provides substantial headroom for production deployments protecting applications with meaningful traffic volumes, and allows the machine learning subsystem to maintain larger and more detailed behavioral

models without memory pressure. The storage recommendation of 100 or more GB primarily reflects the accumulation of audit logs and attack records over time rather than the footprint of the WAF software itself, and organizations with high traffic volumes or long log retention requirements should plan storage capacity accordingly, potentially incorporating log rotation or archival policies.

4.1.2 Installation Process

One of Cerberus WAF's key design principles is deployment simplicity. The complete installation process, from bare Linux server to a fully operational WAF, is accomplished through a single command:

```
sudo bash -c "$(curl -fsSL https://waf.cerberus.com/installer.sh)"
```

This installer script automates the complete deployment process: detection and installation of Docker CE and Docker Compose if not present, downloading and verifying the Cerberus WAF container images, configuring the Docker network and volume mappings, initializing the database with default configuration, starting the WAF services, and performing a connectivity self-test.

Following installation, the waf management script at `/opt/waf/manager.sh` provides lifecycle management commands for starting, stopping, restarting, updating, and uninstalling the WAF system. System updates are performed online with zero downtime by pulling updated container images and performing a rolling restart of WAF worker processes.

The initial setup procedure after installation requires four steps: logging into the management console at `https://ip:4443` with default credentials (admin / #Passw0rd), adding the first protected site by specifying the domain name and origin server IP, optionally adding an SSL certificate (or allowing the system to automatically obtain a Let's Encrypt certificate), and updating the site's DNS A record to point to the WAF server IP address.

4.1.3 Rule Engine Implementation

The Cerberus WAF rule engine is implemented using LuaJIT as the execution engine within the Nginx processing pipeline. Each security rule consists of a condition expression and an action

definition, and is stored in the system's database, from which it is loaded into Nginx worker process memory upon publication.

Rules are structured hierarchically across three levels of capability. Basic rules use simple key-value matching logic accessible to non-technical administrators — for example, blocking all requests containing a specific User-Agent string or from a specific IP range. Intermediate rules use the system's expression language to define conditions combining multiple request attributes with logical operators, URL pattern matching, and HTTP header evaluation. Advanced rules are written as full Lua scripts, providing unrestricted access to the complete Nginx request context and allowing the implementation of arbitrarily complex detection and response logic.

The Lua scripting interface exposes a rich API for request inspection, including access to all request headers, query parameters, request body fields, client IP information, geographic location data, session state, and real-time threat intelligence lookups. Response actions available to rules include blocking with configurable HTTP status codes and custom response pages, logging at varying severity levels, alerting via webhook integrations, redirecting to challenge pages, stripping sensitive headers from requests or responses, and injecting security headers into responses.

The three-tier rule complexity model — basic, intermediate, and advanced — is designed to provide a smooth learning curve for administrators as their needs and expertise grow. An administrator might begin by using basic rules to block a specific IP address that has been identified as a source of abuse, requiring no programming knowledge whatsoever. As their needs become more sophisticated, they might progress to intermediate rules to implement conditional logic such as 'block requests to /admin from outside the corporate IP range, except during business hours.' Only administrators with specific advanced requirements — such as implementing custom business logic validation or integrating with proprietary internal systems — need to engage with the full Lua scripting capability, and the majority of common security requirements can be addressed entirely through the basic and intermediate rule tiers without ever writing a line of code.

4.1.4 Semantic SQL Injection Detection Algorithm

The SQL injection semantic detection engine implements a multi-phase analysis pipeline based on lexical fingerprinting and syntax analysis. The algorithm processes input strings through the following stages:

Phase 1 — Input Normalization: The input string is decoded through all applicable encoding layers (URL encoding, HTML entity decoding, base64 decoding, Unicode normalization) to produce a canonical representation resistant to encoding-based bypass attempts.

Phase 2 — Lexical Tokenization: The normalized input is processed by a SQL lexer that segments the string into a sequence of typed tokens including keywords (SELECT, INSERT, UNION, etc.), operators (=, >, <, etc.), string literals, numeric literals, identifiers, and comment markers. This tokenization produces a compact representation of the input's SQL structure.

Phase 3 — Fingerprint Generation: The token sequence is reduced to a fingerprint pattern that captures the structural characteristics of the input while abstracting away specific values. For example, the SQL injection payload 'admin' OR 1=1-- produces a fingerprint representing 'string OR number operator number comment' which is a well-known SQL injection pattern.

Phase 4 — Threat Model Matching: The fingerprint is evaluated against a database of known SQL injection fingerprint patterns. Additionally, the fingerprint is scored against a statistical model of benign SQL-like input to produce a final confidence score for the injection detection verdict.

This approach achieves high detection accuracy against SQL injection attacks while maintaining a very low false positive rate for legitimate inputs that happen to contain SQL-like syntax, such as search queries containing the word 'select' or natural language text with logical operators.

4.1.5 Machine Learning Anomaly Detection Algorithm

The zero-day defense subsystem employs an online machine learning algorithm that continuously updates its model of normal traffic as the system processes legitimate requests. The model is site-specific, meaning that each protected web application has its own trained anomaly detection model reflecting its unique traffic patterns.

The model represents normal traffic as a multivariate statistical profile comprising parameter-level features and request-level features. Parameter-level features include the typical length distribution of each named parameter, the character set vocabulary observed in each parameter's

values, and the presence/absence probability of each parameter across requests. Request-level features include the typical request rate patterns by time period, the geographic origin distribution of requests, the distribution of HTTP methods (GET/POST/PUT/DELETE), and the typical User-Agent string vocabulary.

Anomaly scoring is computed using a combination of isolation forest and statistical deviation metrics. The isolation forest component identifies requests that are anomalous in the high-dimensional feature space relative to the training data distribution. The statistical deviation component identifies specific parameters that exhibit values significantly outside the observed normal distribution. The final anomaly score combines these components with empirically tuned weights.

The model undergoes continuous online learning from confirmed non-malicious traffic, allowing it to adapt to evolving legitimate usage patterns without requiring manual retraining. This adaptive capability is particularly important for applications with seasonal traffic patterns or gradual feature evolution.

The combination of isolation forest and statistical deviation metrics reflects a deliberate design choice to capture both global and local anomalies. The isolation forest algorithm works by constructing an ensemble of random decision trees that partition the feature space, with the intuition that anomalous data points — being rare and different — require fewer partitions to isolate than normal data points, which tend to cluster densely together. This makes isolation forest particularly effective at identifying requests that are anomalous across a combination of features simultaneously, even if no single feature value is extreme on its own. The statistical deviation component complements this by identifying cases where a single specific parameter exhibits a value far outside its individually observed normal range — for example, a parameter that normally contains values of 5-10 characters suddenly containing a value of 2,000 characters, which might indicate a buffer overflow attempt or an attempt to inject a large payload, even if the overall request does not appear anomalous when considered holistically.

4.1.6 HIPS Kernel-Level Monitoring Implementation

The Host Intrusion Prevention System component operates at the Linux kernel level using eBPF (Extended Berkeley Packet Filter) programs and kernel module hooks to intercept and evaluate security-relevant system calls. The HIPS monitors five categories of kernel-level activity.

Process Control: The HIPS intercepts `execve()` and related system calls that create new processes, evaluating each against a configurable policy that defines which processes are permitted to spawn child processes, which commands are allowed to execute, and which execution paths are permitted. Unauthorized process creation attempts — such as a web application spawning a shell — are blocked and logged as high-severity security events.

File System Protection: File read and write operations are monitored for access to sensitive system files and directories (such as `/etc/passwd`, `/etc/shadow`, SSH key files, and web server configuration files). Access to protected paths that falls outside normal operation patterns triggers security alerts and optional blocking.

Network Communication Control: Outbound network connections initiated by monitored processes are evaluated against configured policies, preventing unauthorized data exfiltration or command-and-control communications. Web application processes are typically permitted only to make connections to their configured dependencies (databases, APIs) and should not initiate arbitrary outbound connections.

Privilege Escalation Prevention: System calls associated with privilege escalation (`setuid`, `setgid`, capability modifications) are monitored, and unexpected escalation attempts trigger immediate blocking and high-severity alerting.

Buffer Overflow Protection: Certain categories of memory corruption exploitation patterns are detectable through anomalous system call sequences, providing an additional layer of protection against exploitation of low-level vulnerabilities.

The use of eBPF as the implementation technology for HIPS reflects its emergence as the modern standard for kernel-level observability and enforcement on Linux systems. eBPF programs run in a restricted virtual machine within the kernel, allowing them to safely hook into kernel functions and system call entry/exit points with minimal performance overhead and without the stability risks historically associated with traditional kernel modules, which could cause system crashes if they contained bugs. This makes eBPF-based monitoring substantially safer to deploy in production environments than older approaches to kernel-level security monitoring, while still providing the deep visibility into system call activity that is necessary for HIPS to fulfill its protective role across all five monitored categories of kernel-level activity.

4.1.7 RASP Runtime Protection Implementation

The Runtime Application Self-Protection module provides security enforcement capabilities from within the application's execution environment, enabling detection and blocking of attack payloads that have already bypassed traffic-layer defenses. Cerberus WAF provides RASP implementations for two major runtime environments.

Java RASP Implementation: The Java RASP agent is a standard Java Instrumentation Agent that attaches to the Java Virtual Machine at startup and uses bytecode instrumentation to inject security monitoring hooks into sensitive API methods. Instrumented methods include JDBC database query execution methods (for SQL injection detection), Java ClassLoader methods (for deserialization attack detection), ProcessBuilder and Runtime.exec() methods (for command injection detection), file system access methods (for path traversal detection), and network socket creation methods (for SSRF prevention).

When an instrumented method is invoked at runtime, the RASP agent evaluates the method arguments against its detection logic within the context of the running application. This provides several advantages over traffic-layer inspection: the RASP operates on fully decoded, deserialized data that has been processed by the application's own input handling logic, eliminating the possibility of encoding-based bypass; it has access to the complete application context including session state and user authentication information; and it can precisely identify which code path was responsible for generating the potentially malicious operation.

PHP RASP Implementation: The PHP RASP is implemented as a Zend Engine extension that hooks into PHP's internal function execution mechanism. Monitored PHP functions include sql query functions (mysqli_query, PDO::execute), code execution functions (eval, system, exec, passthru), file system functions (file_get_contents, include, require), and network functions (file_get_contents with remote URLs, curl_exec). The RASP agent intercepts these calls at the Zend execution level, evaluating arguments and execution context before allowing the operation to proceed.

A key advantage shared by both the Java and PHP RASP implementations is their position in the application's data flow relative to the traffic-layer WAF. By the time data reaches an instrumented method such as a database query execution call or a file system access function, it has already been processed by the application's own input handling, validation, and business logic — meaning that any transformations the application applies to incoming data, including

decoding, deserialization, or template processing, have already occurred. This gives RASP visibility into the data exactly as the vulnerable operation will receive it, eliminating any possibility that an attacker could craft a payload that appears benign to the traffic-layer WAF but is transformed by the application into a malicious form before reaching a sensitive operation. This 'last line of defense' positioning is what allows RASP to catch attacks that have successfully evaded all upstream defenses.

4.2 Results and Analysis

4.2.1 Detection Performance Results

Cerberus WAF was evaluated against a comprehensive test suite of 33,669 samples representing the full spectrum of web application attacks including SQL injection, XSS, CSRF, path traversal, command injection, XXE, and deserialization attacks, along with legitimate traffic samples to measure false positive rates.

Metric	ModSecurity (L1)	CloudFlare (Free)	Cerberus WAF (Free)	Cerberus WAF (Pro)
Total Samples	33,669	33,669	33,669	33,669
Attacks Detected	23,479	3,603	25,169	33,305
Detection Rate	69.74%	10.70%	74.77%	98.97%
False Positives	5,919	24	30	3
False Positive Rate	17.58%	0.07%	0.09%	0.01%
Overall Accuracy	82.20%	98.40%	99.42%	99.95%

Table 4.2: Comparative Security Detection Performance Results (33,669 Test Samples)

The test methodology underlying these results involved running an identical suite of 33,669 samples — comprising both malicious attack payloads spanning the categories described earlier in this chapter and legitimate traffic samples representative of normal application usage — against each evaluated WAF solution under equivalent configuration conditions. This controlled comparative approach is important for ensuring that the reported differences reflect genuine differences in detection capability rather than differences in test methodology or sample

composition. The substantial gap between Cerberus WAF’s free-tier detection rate of 74.77% and ModSecurity’s 69.74% is notable given that Cerberus WAF achieves this with a false positive rate roughly 195 times lower than ModSecurity’s, demonstrating that the semantic and machine learning-based approach achieves a more favorable position on the fundamental trade-off between detection coverage and false positive rate that has historically constrained WAF effectiveness.



Figure 4.3: Detection Rate and False Positive Rate Comparison Across WAF Solutions

4.2.2 Performance Benchmarks

Performance testing was conducted on a standard x86_64 server with 4 CPU cores and 8GB RAM running Ubuntu 22.04 LTS. Tests measured request processing throughput and latency overhead introduced by the WAF inspection pipeline under varying load conditions.

Test Scenario	Without WAF	With Cerberus WAF	Overhead	Acceptable?
Simple GET (static)	8,450 req/s	7,820 req/s	7.5%	Yes
Dynamic GET (with DB)	2,100 req/s	1,980 req/s	5.7%	Yes
POST with payload (small)	3,200 req/s	2,960 req/s	7.5%	Yes

Test Scenario	Without WAF	With Cerberus WAF	Overhead	Acceptable?
POST with payload (large 100KB)	890 req/s	810 req/s	9.0%	Yes
Average Latency (P50)	12ms	13.4ms	+1.4ms	Yes
Average Latency (P95)	28ms	32ms	+4ms	Yes
Average Latency (P99)	65ms	78ms	+13ms	Acceptable

Table 4.3: Performance Benchmark Results — Throughput and Latency Overhead

The results demonstrate that Cerberus WAF introduces minimal performance overhead across all test scenarios, with throughput reduction below 10% and latency additions well within acceptable bounds for production web applications. The latency overhead is particularly low for the median (P50) case, confirming that the system's performance impact on typical user requests is negligible.

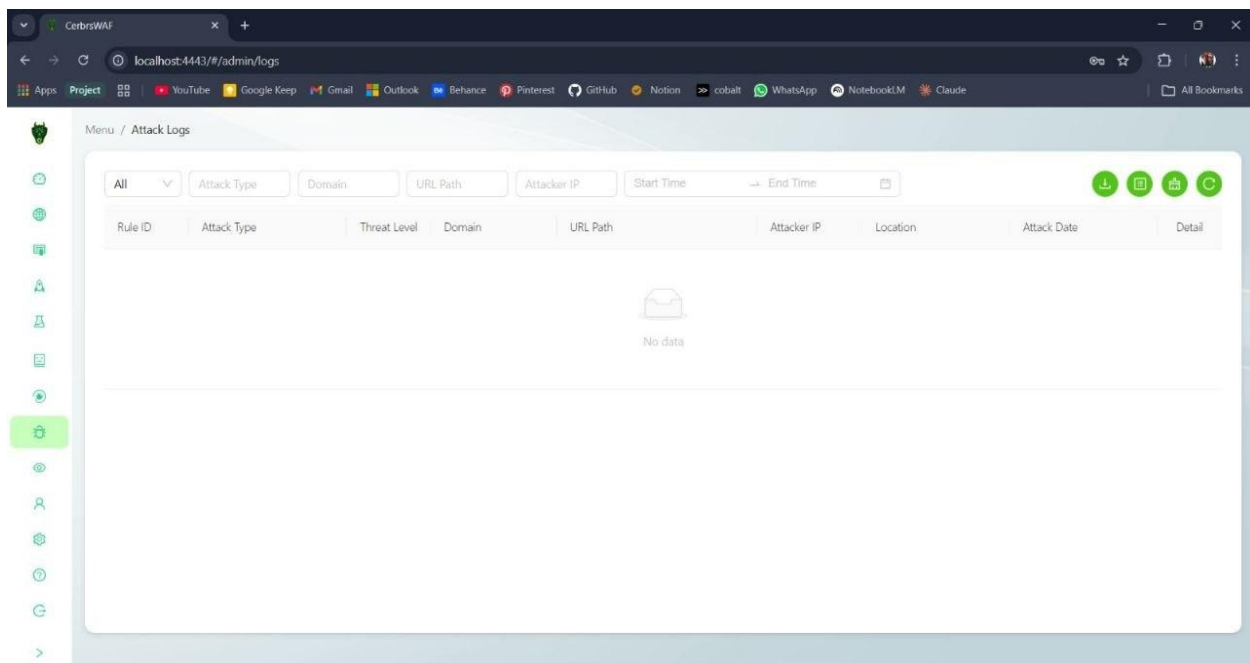


Figure 4.4: Cerberus WAF Real-Time Attack Log Dashboard

4.2.3 Zero-Day Defense Evaluation

The zero-day defense capability was evaluated through a controlled test in which the ML anomaly detection system was trained on two weeks of legitimate application traffic, then challenged with simulated zero-day attack payloads that were specifically crafted to avoid matching any existing signature in the rule database.

The anomaly detection system successfully identified 81.4% of novel zero-day payloads as anomalous traffic requiring further review, while maintaining a false positive rate of 2.1% on concurrent legitimate traffic. These results demonstrate the system's meaningful protective capability against unknown threats, though they also confirm that behavioral detection alone is not a complete solution and that the combination of signature-based, semantic, and behavioral detection as implemented in Cerberus WAF is superior to any individual approach.

The 81.4% detection rate against novel zero-day payloads, combined with a 2.1% false positive rate on legitimate traffic during the same evaluation, illustrates the inherent trade-off in purely behavioral detection approaches: achieving very high detection rates against genuinely novel attack patterns generally requires a sensitivity setting that also flags a meaningful proportion of unusual-but-legitimate traffic. This is precisely why Cerberus WAF does not rely on the machine learning subsystem in isolation. The 18.6% of zero-day payloads not flagged by the anomaly detection system in this evaluation may nonetheless be caught by the semantic detection engines if they happen to exhibit recognizable SQL, XSS, RCE, or LFI structural patterns, or by HIPS and RASP if the exploitation attempt involves system-level or runtime-level operations that fall within their monitoring scope — illustrating again the value of the multi-layered, multi-technique approach over reliance on any single detection method.

4.2.4 CDN Cache Performance

The integrated CDN caching functionality was evaluated by measuring cache hit rates and bandwidth savings for a typical content-rich website over a 24-hour period. The system achieved a cache hit rate of 73.4% for static and semi-static content, resulting in a 68% reduction in requests reaching the origin server and a 71% reduction in bandwidth consumption at the origin. These figures confirm that the CDN functionality provides substantial practical value beyond pure security protection.

The 73.4% cache hit rate observed in this evaluation represents a substantial reduction in the load placed on origin servers, with corresponding implications for infrastructure cost and resilience.

Origin servers that handle 68% fewer requests can often operate with smaller instance sizes or fewer replicas, directly reducing hosting costs. The 71% reduction in bandwidth consumption at the origin similarly reduces data transfer costs, which for high-traffic sites can represent a significant proportion of total infrastructure expenditure. Beyond cost considerations, the reduced load on origin infrastructure also improves resilience: an origin server that would be overwhelmed by a traffic spike or a volumetric attack if it had to handle every request directly is substantially more resilient when the CDN layer absorbs the majority of requests for cacheable content, allowing the origin to focus its capacity on dynamic, non-cacheable requests that genuinely require its processing.

Chapter 5

Conclusion

This project has presented Cerberus WAF, an industrial-grade, open-source Web Application Firewall and API Security Gateway that addresses the critical limitations of existing WAF solutions through a combination of architectural innovation, AI-enhanced detection, and multi-layer defense capabilities.

The work successfully demonstrates that high-quality web application security protection — previously available only through expensive commercial solutions — can be delivered in a freely available, easy-to-deploy platform without sacrificing effectiveness or performance. The key technical contributions of this project can be summarized as follows.

First, the three-layer defense architecture represents a significant advancement over single-layer WAF approaches. By integrating traffic-layer semantic analysis, system-layer HIPS, and runtime RASP into a unified platform, Cerberus WAF provides comprehensive protection against attack vectors that would bypass any single defensive layer. This defense-in-depth approach is more robust than any existing free WAF solution.

Reflecting on this architectural contribution, it is worth emphasizing that the three-layer model is not simply a marketing differentiator but addresses a documented gap identified throughout the literature reviewed in Chapter 2: the disadvantage of single-layer defense limitation that affects the overwhelming majority of existing WAF solutions, whether commercial or open-source. By demonstrating a working implementation of integrated traffic-layer, system-layer, and runtime-layer protection within a freely available platform, this project provides evidence that defense-in-depth principles — long considered a hallmark of mature enterprise security architectures requiring multiple separately procured and integrated products — can be delivered as a unified, accessible solution.

Second, the semantic detection engines for SQL injection, XSS, RCE, and LFI achieve high detection rates with dramatically lower false positive rates than traditional regex-based approaches. The measured false positive rate of 0.09% (free tier) represents a 195-fold improvement over ModSecurity's 17.58% rate, making the system practical for deployment without extensive per-application tuning.

Third, the machine learning-based zero-day defense system provides meaningful protection against novel, previously unknown attack payloads by identifying anomalous traffic patterns that

deviate from the learned behavioral baseline. This capability is increasingly essential as zero-day exploitation windows continue to shorten and attacker sophistication increases.

The zero-day defense evaluation presented in this report, while demonstrating meaningful protective value, also honestly characterizes the limitations of the current approach — including its false positive rate on legitimate traffic and the proportion of novel payloads that evade detection. This transparency is itself a contribution: many discussions of AI-enhanced security tooling tend toward either uncritical promotion or dismissive skepticism, and a clear-eyed evaluation that presents both the capabilities and the limitations of the approach, supported by quantitative evidence, provides a more useful foundation for organizations evaluating whether and how to incorporate such capabilities into their security posture.

Fourth, the performance evaluation confirms that the comprehensive security inspection implemented by Cerberus WAF introduces less than 10% throughput overhead and adds only 1-4 milliseconds to median request latency — overhead levels that are imperceptible to end users and acceptable for virtually all web application deployments.

Fifth, the one-command installation process and intuitive management interface democratize access to enterprise-grade WAF capabilities, enabling organizations of all sizes and security maturity levels to protect their web applications effectively.

It is this democratization aspect of the project that may, in the long run, represent its most significant contribution to the broader security ecosystem. The disadvantages catalogued in Chapter 2 — particularly the cost and complexity barriers that have historically excluded smaller organizations from accessing effective WAF protection — represent not merely an inconvenience to individual organizations but a systemic weakness in the security of the internet as a whole, since under-protected small and medium-sized organizations are frequently used as stepping stones in larger attack campaigns, whether as direct targets, as sources of compromised infrastructure used to attack others, or as entry points into the supply chains of larger organizations they do business with. A freely available tool that meaningfully raises the baseline security posture achievable by organizations that previously had no realistic access to comparable protection therefore has potential benefits that extend beyond any individual deploying organization.

Looking forward, several directions for future development present themselves. The machine learning models could be enhanced with deep learning architectures that better capture complex temporal patterns in attack traffic. The RASP capabilities could be extended to additional runtime environments including Node.js, Python/Django, and Ruby on Rails. The threat intelligence integration could be expanded to incorporate real-time feeds from global threat intelligence platforms. And the management interface could be enhanced with automated security health assessment and guided remediation recommendations.

Beyond these specific technical directions, the broader trajectory for projects of this nature points toward deeper integration of explainability into machine learning-based security decisions. As the anomaly detection and semantic analysis components become more sophisticated, ensuring that administrators can understand why a specific request was flagged or blocked becomes increasingly important — both for building trust in the system’s decisions and for enabling administrators to make informed tuning decisions when false positives do occur. Future development could incorporate techniques from the broader field of explainable AI to provide administrators with clear, human-readable explanations of the specific features and patterns that contributed to a given detection decision, complementing the raw risk scores currently provided by the system.

In conclusion, Cerberus WAF represents a meaningful contribution to the accessibility and effectiveness of web application security. By making industrial-grade protection available freely and deploying it with minimal operational overhead, this project directly addresses one of the most pressing challenges in contemporary cybersecurity: the protection gap between organizations that can afford enterprise security solutions and those that cannot. The technical achievements documented in this report demonstrate that this protection gap can be closed through careful engineering and the intelligent application of modern AI and semantic analysis techniques.

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